Math Bootcamp

Introduction

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Aim of this course

- The purpose of the bootcamp is to go through some of the mathematical skills required for the core economics courses. We will emphasize:
  - technique (e.g., how to solve an optimization problem);
  - problem solving; and
  - application to economics.
- Self guided
- 3 modules
  - Linear Algebra
  - Calculus and optimization
  - Probability and statistics

Why these?

- Economic models are really just sets of simultaneous equations, that are best solved using matricies (linear algebra):
  - E.g., Demand and supply as a function of price (Equilibrium D=S)
    - Behavioral equations
      - $Q_1^D = a - b.P_1 + c.P_2$
      - $Q_2^D = d - e.P_2 + f.P_1$
    - Equilibrium equations
      - $Q_1^D = Q_1^S$
      - $Q_2^D = Q_2^S$
    - Exogenous variables and Parameters
      - Endogenous (determined by model): Q_1^D, Q_2^D, P_1, P_2
      - Exogenous (fixed) variables: Q_1^S and Q_2^S
      - Parameters a, b, c, d, e, f – real numbers $\mathbb{R}$

Why these?

- The agents in our models are always trying to optimize (calculus)
  - In deciding what to buy, consumers try to maximize happiness (utility) subject to their income
  - In deciding what and how to produce goods, firms try to maximize their profits
- Ex-post econometric analysis is founded on probability and statistics

Types of numbers

- Natural – arise in counting (1,2,3…..)
- Integers – adds negative numbers to natural numbers (….-3,-2,-1,0,2,3…..)
- Rational – adds to integers, any number that is the result of a division of integers (except 0)
- Irrational – other numbers that cannot be expressed as ratios of integers (e.g. $\sqrt[2]{2}$)
- Real ($\mathbb{R}$) – all rational and irrational

Properties of real numbers

- a, b $\in \mathbb{R}$, then
  - a + b $\in \mathbb{R}$ and
  - a.b $\in \mathbb{R}$
- a + b = b + a
- a + (b+c) = (a+b) + c
- a.(b+c) = a.b + a.c
- a + 0 = a
- a.0 = 0
- 1.a = a
- a – a = 0
- a.(1/a) = a/a = 1

Variables

- Use x_i to signify some variable of interest
  - *x* is the variable of interest quantity demanded or produced, price, value, employment, profits………..
  - i signals that there is a value for these variables for a set of items (i). For instance, i could represent a set of firms producing the good, the set of commodities, countries, time period….. for which we have values for this variable.
- Sum operator:
  - to sum these variables over i (from 1..n)
    - $\sum_{i=1}^{n} x_i = x_1 + x_2 + x_i \ldots + x_n$
  - In our example?
    - $\sum_{i=1}^{3} x_i = x_1 + x_2 + x_3 = 1.2 + 0.9 + 1.5 = 3.6$

Firm i	X_i	Value of X_i Production (millions)
1	X_1	1.2
2	X_2	0.9
3	X_3	1.5

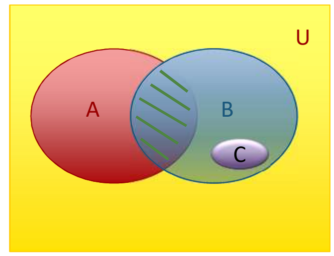
Variables and sum operator

- Other things to note about the sum operator
  - $\sum_{i=1}^{n} c = n.c$, where c is a constant
  - $\sum_{i=1}^{n} c.x_i = c.\sum_{i=1}^{n} x_i$
  - $\sum_{i=1}^{n} (a.x_i + b.y_i) = a.\sum_{i=1}^{n} x_i + b.\sum_{i=1}^{n} y_i$, where a and b are constants
  - $\sum_{i=1}^{n} \frac{x_i}{y_i} = \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} y_i}$
- Can you show that: $\sum_{i=1}^{n} (x_i - \bar{x}) = 0$?
  - Where $\bar{x}$ is the mean and is given by: $\bar{x} = \frac{1}{n}.\sum_{i=1}^{n} x_i$

$$\sum_{i=1}^{n} (x_i - \bar{x}) = \sum_{i=1}^{n} x_i - n.\bar{x} \quad = \sum_{i=1}^{n} x_i - \cancel{n}.\frac{1}{\cancel{n}}.\sum_{i=1}^{n} x_i = 0$$

Sets

- Universal set (U)
- A, B and C are subsets of U
- C is also a subset of B
- Union $(A \cup B)$ (red and blue together)
- Intersection $(A \cap B)$ (green hashed area)
- Empty set (null set ∅)
- Complement $\bar{B}$ (everything outside of B)



Venn diagram

Dimensions

- Important in economics
  - dollars, dollars per kg
  - Cannot add applies and oranges
  - All dimensions in an equation must match
    - Profit (\$) = revenue (\$) – costs (\$)
    - Revenue (dollars) ≠ Price $\left(\frac{\$}{pound}\right)$ = Marginal revenue $\left(\frac{\$}{pound}\right)$
    - Revenue (dollars) = Price (dollars per item). Quantity (number of items)
      - $\$ = \frac{\$}{\cancel{pound}} . \cancel{pounds}$

Percent and proportionate changes

- A lot of variables used in economics are reported in percentages (interest rates, tax rates, inflation rates, unemployment rates…)
  - To convert into a proportion:
    - Proportion = (Percent rate)/100
  - To convert to values
    - Proportion * Relevant Value
  - e.g., 10% unemployment = 10/100 * 156 million (labor force) = 15.6 million people unemployed.

- Proportionate changes in a variable over time

$$\hat{x} = \frac{\Delta x}{x_o} = \frac{(X_1 - X_0)}{(X_0)}$$

  - Where: x_1 is the final value and x_o the initial value
  - Convert to percent change by multiplying by 100: $\hat{x}.100$

- Percentage point change → changes in rates:
  - For example: unemployment rate rises from 5% to 10%.
  - Percentage point change = 10 - 5 = 5 percentage points

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Functions

- Function is a rule that associates each element of x with one element of y ($y = f(x)$ or $f: y \rightarrow x$)
  - x co-domain (independent variable)
  - y domain (dependent variable)
  - $x = f^{-1}(y)$ (inverse of function)
  - Or write implicitly as $f(x, y) = 0$

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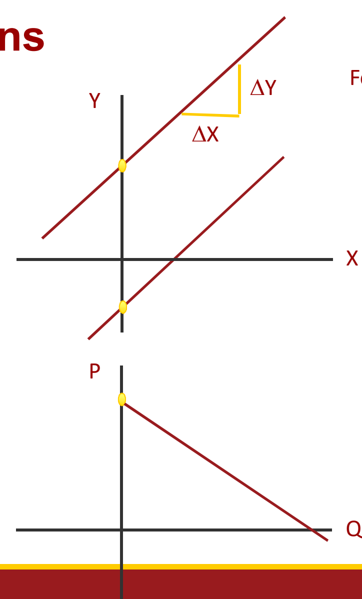
Linear functions

- Examples of linear equations
  - $a + b.X = Y$
  - $a + b.X + c.Z + d.W = Y$
- Examples of non-linear equation
  - $x^2 + y^2 = z$
  - $x^{1/2} = y$
  - $\beta_0 + \beta_1.X + \beta_2.X^2 = Y$
- Conditions for linear equations
  - have only powers of 1
  - a, b, c must be real numbers

Graphing simple linear equations

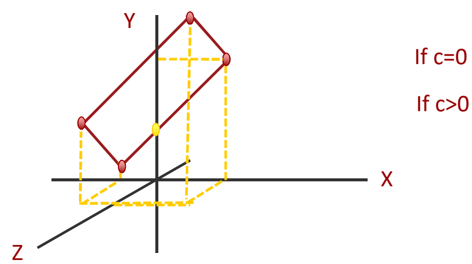
- Draw $\mathbb{R}^2$
  - $Y = a + b.X$
    - Y-intercept is a
    - slope is b (marginal effect: ΔY/ΔX = b)
  - Example: demand curve
    - $Q = a - b.P$
    - $P = \frac{a}{b} - \frac{1}{b}.Q$



For a given ΔX,
$\Delta Y = b.\Delta X$
$b = \frac{\Delta Y}{\Delta X}$

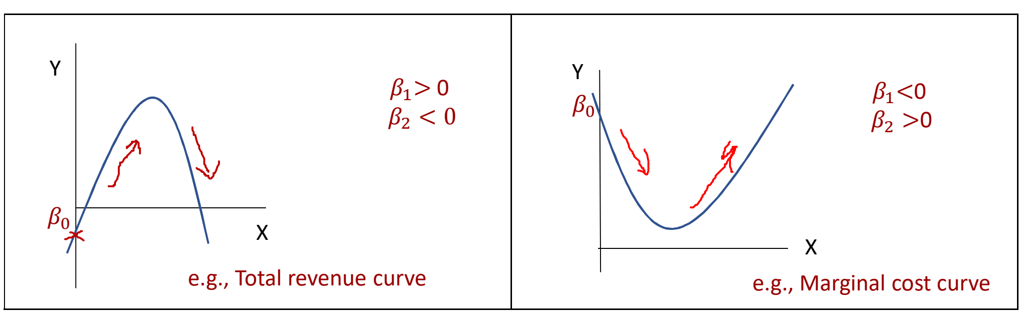
Graphing simple linear equations

- Draw $\mathbb{R}^3$
  - $Y = a + b.X$
    - Y-intercept is a
    - slope is b (marginal effect: ΔY/ΔX = b)
  - $Y = a + b.X + c.Z$
    - partial effects if multiple dimensions: ΔY/ΔX = b, assuming Z is unchanged (ceteris paribus)



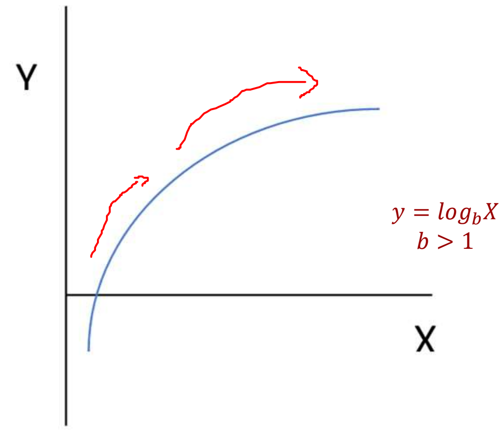
Quadratics

- Quadratic functions
  - $Y = \beta_0 + \beta_1.X + \beta_2.X^2$



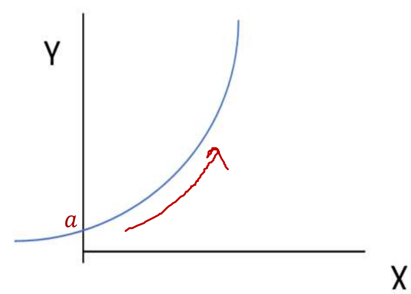
Logarithmic

- Y = log(X)
  - Log(X) < 0, 0<x<1
  - Log(1) = 0
  - Log(X) >0, X>1
  - $X = b^y$ then $y = log_b X$
  - $b = e$ then $y = log_e X$
- Diminishing marginal returns
- Some other rules for logs
  - Log(XY) = log(X) + log(Y)
  - Log(X/Y) = log(X) – log(Y)
  - $\mathrm{Log}(X^c) = \mathrm{c.Log}(X)$
  - Log(X1) – log(X0) = (X1 – X0)/X0 = ΔX/X0 = $\hat{X}$ (proportionate change)



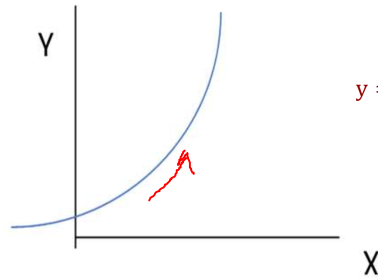
Exponential functions

- $\mathrm{y} = \exp(\mathrm{x})$ or $y = a.b^x$
- Increasing returns
- Notes on exponentials
  - Exp(0) = 1
  - Exp(1) = 2.7183
  - Log(exp(X)) = X
  - Exp(Log(X)) = X
  - exp(X+Y) = exp(X).exp(Y)
  - exp(c.log(X))=x^c

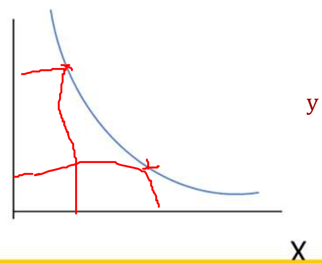


Power Functions

- Power function: $y = a.x^b$
  - E.g., Cobb Douglas $y = x^a.z^b$
- Rectangular Hyperbola $y = \frac{1}{x} = x^{-1}$
  - Consumers preferences for two goods X and Y – diminishing marginal utility ($\mathbb{R}_+$)
- Notes on powers
  - $a^n.a^m = a^{n+m}$
  - $(a^n)^m = a^{n.m}$
  - $\frac{a^n}{a^m} = a^{n-m}$
  - $\frac{a^n}{a^n} = a^0 = 1$

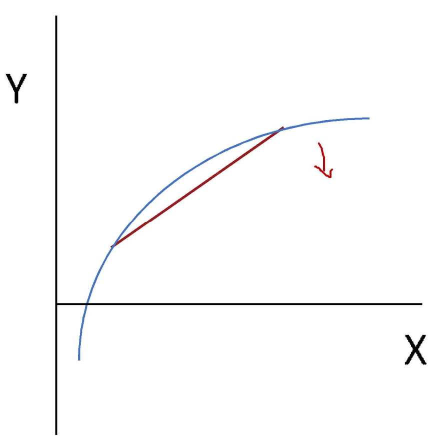


$y = a.x^b$
b>1

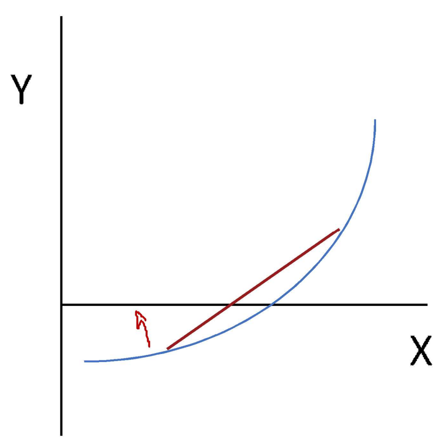


$y = a.x^b$
b<-1

Concave



Convex



Non-linear functions: Circles

- $X^2 + Y^2 = a$
  - where a is the radius squared r^2
- More generally: $\beta_1 . X^2 + \beta_2 . Y^2 = a$
  - When parameters $\beta_1 = \beta_2$ you get a circle
  - When parameters $\beta_1 \neq \beta_2$ differ you get an ellipse
- When you add a third dimension you get a sphere
  - $\beta_1 . X^2 + \beta_2 . Y^2 + \beta_3 . W^2 = a$